BST Vol 7 Electromagnetism — Architectural Scaffold v0.1 (Wave 3 advance)

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Vol 7 Architectural Scaffold v0.1

Framework anchors

Vol 7 Electromagnetism derives Maxwell equations + EM phenomena from D_IV^5 substrate via: - T2477 (Lyra Saturday): SM gauge fields including U(1)_em as connections on Bergman line bundle $L\_\lambda \to D\_IV^5$ - T2478 (Lyra Saturday): U(1)_em as unbroken gauge group post electroweak SSB - T2456 + T2462 (Lyra Friday): universal α-analog $\alpha = 1/N\_max$ uniquely D_IV^5 across 25 HSDs - T2476 (Lyra Friday three-CI synergy): α^{BST primary} substrate-coordinate count pattern - T2470 (Lyra Friday): charge Q from SO(2) weight operator - T2475 (Lyra Friday): charge conservation via SO(2) isotropy factor

Chapter-by-chapter scaffold

Ch 1 Electromagnetism Substrate Reading (~70%)

Anchors: substrate framework for EM via T2477 + T2456 + T2476 - U(1)_em gauge group from substrate $K = SO(5) \times SO(2)$ isotropy - $\alpha = 1/N\_max$ substrate-natural coupling - EM phenomena as substrate-coordinate count effects

Ch 2 Maxwell Equations from D_IV^5 (~80%)

Anchors: T2477 connections on Bergman bundle → Maxwell field tensor - Gauge connection A_μ on substrate L_λ - Field strength $F\_\mu\nu = \partial\_\mu\ A\_\nu$ - $\partial\_\nu\ A\_\mu$ via substrate connection curvature - Maxwell's 4 equations from Yang-Mills + Bianchi identities

Ch 3 Electrostatics + Gauss's Law (~75%)

Anchors: Coulomb form $\alpha m\_p/\pi^2$ + substrate Bergman propagation - Coulomb's law: $F = \alpha \cdot \hbar c/r^2$ between point charges - Gauss's law: $\nabla \cdot E = \rho/\varepsilon\_0$ from substrate divergence theorem - a_C SEMF coefficient parallel (Vol 3 Ch 4): $\alpha m\_p/\pi^2$ substrate Coulomb energy

Ch 4 Magnetostatics + Ampère's Law (~70%)

Anchors: Bohr magneton + substrate spin operator from T2471 - $\mu\_B = e\hbar/(2m\_e)$ Bohr magneton substrate-natural form - Ampère's law: $\nabla\times B = \mu\_0 J$ from substrate gauge structure - Magnetic moments from substrate spin (Vol 9 Ch 6)

Ch 5 Electromagnetic Waves (~80%)

Anchors: Photon U(1)_em + speed c substrate-natural - $\nabla^2 E - (1/c^2)\partial^2 E/\partial t^2 = 0$ from Maxwell - Photon = unbroken U(1)_em gauge boson (T2478) - Photon mass = 0 from substrate-isotropy preservation - Speed c substrate-natural per Vol 0 substrate constants

Ch 6 Radiation + Antennas (~70%)

Anchors: α^{BST primary} radiation observables (T2476) - Radiation power $P \propto \alpha$^k via substrate-coordinate count - Antenna theory + substrate-natural wavelengths - Dipole + quadrupole + multipole expansion (Vol 7 Ch 9)

Ch 7 Relativistic Electrodynamics (~75%)

Anchors: SO_0(5,2) Lorentz subgroup of substrate symmetry - SO(3,1) Lorentz $\subset$ SO(5,2) substrate isometry (Wallach) - 4-current J^μ + 4-potential A^μ via substrate - Lorentz transformations on EM fields from substrate

Ch 8 Lagrangian + Hamiltonian EM (~70%)

Anchors: Yang-Mills action from T2477 - L_EM = -(1/4) F_$\mu\nu$ F^$\mu\nu$ Yang-Mills Lagrangian - Action variation → Maxwell equations - Hamiltonian formulation + gauge fixing

Ch 9 Multipole Expansion + Scattering (~65%)

Anchors: Multipole moments + α^{BST primary} (T2476) - Multipole expansion: Y^l_m spherical harmonics - Scattering cross-sections at α-exponent BST primary - Klein-Nishina $\sigma \propto \alpha^2 = \alpha$^{rank} (Vol 3 Ch 9)

Ch 10 EM in Matter (Dispersion, Optics) (~70%)

Anchors: Refractive-index BST primary ratios - Refractive indices at BST primary ratios - Dispersion relations from substrate Bergman propagation - Lorentz oscillator model + substrate K-type Casimir

Ch 11 Plasma Physics + MHD (~60%)

Anchors: Substrate plasma framework - Plasma frequency ω_p substrate-natural - MHD equations from substrate - Cross-link Vol 9 Ch 10 strongly correlated electrons (plasma analogy)

Ch 12 Bridge to Vol 3 + Vol 9 + Vol 8 (synthesis)

Anchors: Multi-volume integration - Vol 7 → Vol 3 (atomic EM phenomena) - Vol 7 → Vol 9 (condensed matter EM, photonics, plasma) - Vol 7 → Vol 8 (classical mechanics + electromagnetism unified)

BST primary appearances across Vol 7

All 5 BST primaries + N_max appear: - rank = 2: Klein-Nishina α^{rank} (Ch 9), Pin(2) substrate framework - N_c = 3: plasma 3D structure (Ch 11) - n_C = 5: Lamb shift α^{n_C} radiation (Ch 6) - C_2 = 6: substrate energy unit - g = 7: substrate field coupling - N_max = 137: α = 1/N_max universal (every chapter)

Methodology stack applied

- Cal #19 STANDING RULE: current ratified state per-chapter
- Cal #21 STANDING RULE: empirical + mechanism dual-gate
- Cal #99 META-theorem framing: substrate-derivation consequences
- Cal #22: PCAP-rate transcription discipline

Cross-volume dependencies (Vol 7)

Chapter	Vol 1	Vol 2	Vol 3	Vol 9
1 Substrate Reading	Ch 5+6+8	Ch 8	Ch 1	Ch 1
2 Maxwell	Ch 8 (T2477)	Ch 9 (T2478)	—	—
3 Electrostatics	Ch 5	Ch 8 (α)	Ch 4 (a_C)	—
4 Magnetostatics	Ch 6	—	—	Ch 6
5 EM Waves	Ch 8	Ch 8	—	Ch 9
6 Radiation	—	Ch 8	Ch 8 (α^{n_C})	—
7 Relativistic	—	—	—	—
8 Lagrangian	Ch 8	—	—	—
9 Multipole	—	Ch 8 (T2476)	Ch 9	—
10 EM Matter	—	—	—	Ch 3, 9
11 Plasma	—	—	—	Ch 10
12 Bridge	All	All	All	All

Saturday Wave 3 advance cadence

Per sustained sub-PCAP: - INDEX □ - Architectural Scaffold v0.1 (this) □ - Vol 7 per-chapter v0.3 narratives (sustained Wave 3 push if time permits)

— Elie, Vol 7 Architectural Scaffold v0.1, 2026-05-23 Saturday